ANALYSTLAB AFRICA

Data Science Internship Programme

Heart Disease Prediction
Data Preprocessing Report

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Project: Heart Disease Prediction
Dataset: Heart Disease Prediction Dataset

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1. Business Problem

The primary objective of this project is to prepare a dataset for machine learning to accurately predict heart disease. This involves a comprehensive series of data inspection, cleaning, feature engineering, encoding, scaling, outlier treatment, and feature selection steps. The ultimate goal is to build a robust predictive model that can identify individuals at risk of heart disease, thereby aiding in early diagnosis and intervention.

2. Preprocessing Decisions and Justifications

2.1. Data Inspection and Initial Assessment

Data Loading and Shape: The dataset, heart.csv, was loaded into a pandas DataFrame. It initially consisted of 918 rows and 12 columns, providing a substantial number of observations for analysis.

Data Types: Initial inspection revealed continuous numerical columns (Age, RestingBP, Cholesterol, MaxHR, Oldpeak, HeartDisease), a binary numerical column (FastingBS), and categorical columns (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope). RestingBP and Cholesterol were identified as integers but contained anomalous '0' values, which were treated as missing. FastingBS is a binary feature (0 or 1) rather than a continuous variable, and was therefore treated separately from the continuous numerical features during outlier detection.

Missing Values: An initial check for NaN values reported no missing data. However, a deeper dive into the numerical columns (RestingBP and Cholesterol) revealed several entries with a value of 0. These 0 values are physiologically impossible for blood pressure and cholesterol levels, indicating erroneous or missing data.

Justification: Treating these 0s as actual values would distort statistical analyses and model training. Therefore, these 0s were explicitly replaced with np.nan to accurately represent them as missing entries. Subsequently, these NaN values were imputed using the *median* of their respective columns. The median was chosen over the mean to minimise the impact of any remaining extreme values or skewed distributions, providing a more robust central tendency for imputation when data are not missing completely at random (Little & Rubin, 2019).

- Cholesterol: 172 records with 0 values were replaced with np.nan and then imputed with the median cholesterol level of 237.0.
- RestingBP: 1 record with a 0 value was replaced with np.nan and then imputed with the median resting blood pressure of 130.0.

Duplicate Records: No duplicate rows were found in the dataset (df.duplicated().sum() returned 0). This ensured that each observation contributed unique information to the analysis.

2.2. Outlier Detection and Handling

Method: The Interquartile Range (IQR) method was used to identify potential outliers in the continuous numerical features MaxHR and Oldpeak (Tukey, 1977). FastingBS was excluded because it is a binary feature rather than a continuous variable. Boxplots were first used to visually inspect the distributions of MaxHR and Oldpeak before the IQR boundaries were calculated. Outlier detection and capping were carried out on the original, pre-scaling measurement units, so that the IQR boundaries would remain clinically interpretable. The boundaries were computed as follows:

$$IQR = Q3 - Q1$$

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

where Q1 and Q3 represent the 25th and 75th percentiles of the feature distribution, respectively (Tukey, 1977). Any value falling below the lower bound or above the upper bound was flagged as a potential outlier and capped at the corresponding boundary.

IQR Boundaries in Original Units (used for capping)

- MaxHR (original units): Q1 = 120.00, Q3 = 156.00, IQR = 36.00, lower bound = 66.00, upper bound = 210.00.
- Oldpeak (original units): Q1 = 0.00, Q3 = 1.50, IQR = 1.50, lower bound = −2.25, upper bound = 3.75.

These boundaries were calculated in the original clinical measurement units, before any scaling was applied, so that the natural range of each variable could be interpreted directly.

Justification: Values falling outside these boundaries were capped at the calculated lower and upper bounds rather than removed (Winsorisation), preserving all observations for subsequent analysis and model training while reducing the influence of extreme values (Dixon, 1960). This approach was preferred over simply removing the rows to ensure that no valuable information from potentially rare but legitimate observations was lost.

Outlier treatment was completed before feature scaling. Once MaxHR and Oldpeak had been capped in their original units, StandardScaler was applied to these two columns, producing standardised versions of each feature with a mean of 0 and a standard deviation of 1.

Observations:

- FastingBS: Confirmed as a binary feature (0 or 1) rather than a continuous variable, and therefore required no IQR-based treatment.
- MaxHR: Displayed a fairly symmetrical distribution with some values outside the whiskers. The calculated lower bound (66.00) was above the observed minimum of 60, so low values were capped upward to 66.00; the calculated upper bound (210.00) exceeded the observed maximum of 202, so no values required capping at the upper end.
- Oldpeak: Showed a right-skewed distribution with several high outliers and a few low outliers. Both boundaries were within the observed range, so values above 3.75 were capped downward and values below −2.25 were capped upward.

3. Feature Engineering Techniques

3.1. Creating New Features

Two new categorical features were engineered to capture additional insights:

- Age_Group: Categorized Age into Young (0-44), Middle-Aged (45-59), and Elderly (60+).
- BP_Category: Categorized RestingBP into Normal (0-119), Elevated (120-138), and Hypertension (139+).

Justification: Creating these aggregate categorical features can help models identify patterns that might not be obvious from continuous numerical values alone. They provide a more human-interpretable representation of these physiological measurements, potentially simplifying decision boundaries for certain models.

3.2. Feature Encoding

Method: All nominal categorical columns (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope, Age_Group, BP_Category) were converted into numerical format using One-Hot Encoding, with the first category of each variable dropped (drop_first=True) to avoid redundant columns and reduce multicollinearity among the encoded features (Potdar et al., 2017).

Justification: One-Hot Encoding was chosen to transform categorical variables into a format suitable for machine learning algorithms. This method creates binary (0 or 1) columns for each category, preventing the model from assuming an artificial ordinal relationship between categories, which would happen with label encoding for nominal variables (Potdar et al., 2017). It also avoids misleading magnitude interpretations, ensuring that each category is treated independently and equally by the model. The encoding was implemented using standard preprocessing utilities widely adopted in machine learning workflows (Pedregosa et al., 2011).

3.3. Feature Scaling

Method: The numerical features MaxHR and Oldpeak were scaled using StandardScaler (Pedregosa et al., 2011).

Justification: Standardisation (Standard Scaling) transforms the data to have a mean of 0 and a standard deviation of 1. This is crucial for many machine learning algorithms, especially those that rely on distance calculations, like K-Nearest Neighbours, or gradient descent-based algorithms, like Support Vector Machines and Neural Networks, to ensure that features with larger numerical ranges do not disproportionately influence the model's performance. By scaling, all features contribute equally to the distance metrics and optimisation processes.

4. Key Observations from Exploratory Data Analysis (EDA)

Univariate Analysis:

- HeartDisease: The target variable is slightly imbalanced, with approximately 55% of patients having heart disease and 45% not having it.
- Age: Follows a moderate distribution, with a mean of around 53.5 years.
- FastingBS: Mostly patients with fasting blood sugar ≤ 120 mg/dl (704 individuals) compared to those > 120 mg/dl (214 individuals).
- Sex: The dataset is predominantly male (725 males vs. 193 females), indicating a demographic imbalance.
- ChestPainType: 'ASY' (Asymptomatic) is the most common type, highlighting that many patients might have heart disease without typical chest pain.
- RestingECG: 'Normal' readings are most prevalent.
- ExerciseAngina: Most patients do not experience exercise-induced angina.
- ST_Slope: 'Flat' is the most common ST segment slope.

Bivariate Analysis and Feature Correlations:

- Age vs. MaxHR: An inverse relationship exists, where MaxHR tends to decrease with age.
- Age vs. Oldpeak: Older patients tend to exhibit higher Oldpeak values.
- MaxHR vs. Oldpeak: An inverse relationship, lower MaxHR is associated with higher Oldpeak.
- HeartDisease with Numerical Features: Patients with heart disease are generally older, have lower MaxHR, and higher Oldpeak. RestingBP and Cholesterol showed weaker correlations after cleaning.
- HeartDisease with Categorical Features (after encoding):
- ST_Slope_Flat: Strongest positive correlation (0.55) with heart disease.
- ST_Slope_Up: Strongest negative correlation (-0.62) with heart disease, indicating its presence strongly suggests no heart disease.
- ExerciseAngina_Y: Strong positive correlation (0.49).
- Oldpeak: Positive correlation (0.41).
- MaxHR: Negative correlation (-0.40).

- ChestPainType_ATA: Strong negative correlation (-0.40).
- Sex_M: Positive correlation (0.31).
- FastingBS=1: Higher prevalence in heart disease patients.
- Age_Group_Elderly and BP_Category_Hypertension: Show positive correlations with heart disease.

5. Conclusion: Final Prepared Dataset

After comprehensive data preprocessing, the dataset clean_encoded is now prepared for machine learning. The steps undertaken include:

- Handling anomalous '0' values in RestingBP and Cholesterol through median imputation.
- Confirming no duplicate records.
- Creating new, more informative categorical features (Age_Group, BP_Category).
- Transforming all nominal categorical variables into a numerical format using One-Hot Encoding.
- Scaling continuous numerical features (MaxHR, Oldpeak) using StandardScaler.
- Treating outliers in MaxHR and Oldpeak using the IQR-based capping method, applied in the original measurement units before feature scaling, to ensure robust model training.

The final dataset, clean_encoded, contains 918 entries across 20 columns (including the target variable and newly created/encoded features), with all features being numerical and appropriately scaled or encoded. This dataset is now ready for the next phase of machine learning, including model training, evaluation, and further optimisation, leveraging the insights gained from the preprocessing stage to build an accurate heart disease prediction model.

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